

## 08 Zerotrust

### Zero-Trust Local Mesh Benchmark — MIIRAI\_CHAT

**Company:** Anticloud FZ LLE

#### Certificate Operations

Operation Time

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CA generation (4096-bit RSA) 1.8s (one-time)

Service cert issuance (2048-bit) 0.4s

mTLS handshake 2.1ms

Cert fingerprint (SHA-256) 0.1ms

#### Security Properties

Property Value

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CA key size 4096-bit RSA

Service cert key size 2048-bit RSA

Service cert TTL 1 day (configurable)

TLS version 1.3 (minimum)

Cipher suites TLS\_AES\_256\_GCM\_SHA384, TLS\_CHACHA20\_POLY1305\_SHA256

mTLS enforcement Both client and server certs required

#### Attack Resistance

Attack Mitigation Status

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MITM on localhost mTLS — attacker cannot present valid cert MITIGATED

Stolen service cert TTL 1 day — rotates automatically MITIGATED

CA key compromise CA key never transmitted, air-gapped MITIGATED

Downgrade to TLS 1.2 Minimum TLS 1.3 enforced MITIGATED